Fiscal Year:	FY 2024	**Task Last Updated:**	FY 02/01/2024
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Project Title:	Solution Convection and the Nucleation Precursors in Protein Condensation		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Cross-Element Discipline:	None		
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Zip Code:	77204-4004	**Congressional District:**	18
Comments:			
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:			
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Proteins form a variety of solid condensates in their native environments: crystals, amyloid fibrils, amorphous agglomerates, and others. These condensates play essential roles in the physiology of the host organisms, in the progression of numerous diseases, and in numerous industrial processes. Some of these condensates, such as crystals and sickle cell hemoglobin polymers, are ordered; and the protein molecules retain their native conformation. In amyloid fibrils, another ordered condensate, the proteins refold into a beta-sheet-rich structure. By contrast, amorphous aggregates consist of fully disordered protein chains. The formation of solid protein condensates divides into two stages: nucleation of new solid domains, followed by their growth. Nucleation governs how many condensates appear in a certain volume and how they distribute in space and time. In this way, nucleation controls some of the crucial properties of the emerging populations of condensates. Nucleation is by far more complex than growth, as it comprises not only the creation of new bonds in the condensates but also a search through the space of structural parameters of a stable

Task Description:	molecular arrangement. Not surprisingly, theoretical and experimental studies of nucleation have often produced results that appear inconsistent and even contradictory. A recent breakthrough declared that the nucleation of protein ordered and disordered solids may be hosted by mesoscopic protein-rich precursors, sometimes called clusters. The role of the nucleation precursors and their responses to environmental variables elucidated numerous previously elusive features of protein condensation. It was established that the ability to control the population of the mesoscopic solute-rich clusters affords the opportunity to manipulate the rate of solids nucleation. The proposed research focusses on protein solid condensates and aims to understand how moderate and weak solution flows, common in live tissues and a frequent element of industrial procedures, affect their nucleation. Solution shear, known to subtly modify protein conformation even at low levels, may impact the properties of the populations' mesoscopic protein-rich clusters and, in turn, strongly enhance or suppress the rate of crystal nucleation. Our objective is to establish whether, and how, moderate and weak solution flows enhance or suppress the populations of mesoscopic precursors that host the nucleation of protein crystals, amyloid fibrils, and amorphous agglomerates. The main tool to address how solution shear flow may affect the properties of the nucleation precursors is the huge array of data collected in the Light Microscopy Module Biophysics 2,5 (LMMBIO 2,5) experiment. In LMMBIO 2,5 data were obtained with the proteins p53, hnRNPA2, and lysozyme. The p53 clusters are essential precursors for the nucleation of p53 amyloid fibrils that are a part of cancers' pathology; hnRNPA2 clusters evolve into disordered agglomerates, which relate to amyotrophic lateral sclerosis (ALS, also known as Lou Gehrig's disease); whereas mesoscopic lysozyme-rich clusters host the nucleation of lysozyme crystals. The effects of solution shear flow will become apparent from comparing the results of the extremely valuable space experiments, in which most solution flows are suppressed, to duplicates in the laboratory where buoyancy-driven convection dominates solute transport. Successful completion of the proposed research based on the data obtained in the LMMBIO 2,5 experiments will provide fundamental insight on how shear flow impacts the precursors for the nucleation of protein crystals, amyloid fibrils, and amorphous agglomerates. The expected results will illuminate the role of weak and moderate solutions flows in live tissues on the formation of physiological and pathological protein condensates and guide the optimization of protocols in the chemical and pharmaceutical industries.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Successful completion of the proposed research will provide understanding and allow manipulation of nucleation, polymorphism, and structure selection of numerous natural and engineered protein condensates by solution shear flow. As an additional outcome, the expected results will provide understanding why protein crystallization experiments in microgravity have often generated unexpectedly low or high numbers of crystals.
Task Progress:	New Project for FY24
Bibliography Type:	Description: (Last Updated: 12/05/2024)